
Steps to Run the Program in Custom Mode

1. Run the Program

Open a terminal in the project directory and execute the following command:

```
python main.py
```

2. Select Custom Network Mode

When prompted:

Choice (1/2):

Enter:

1

This selects **Custom Network Mode**.

3. Enter Directed Edges

At the prompt:

edge>

Enter directed edges in the format:

from_node to_node capacity

- Enter **one edge per line**
- Type done after entering all edges

Example:

edge> Source J1 12

edge> Source J2 10

edge> J1 HOSP 8

edge> J2 RES 7

edge> HOSP Sink 8

edge> RES Sink 7

edge> done

4. Enter Consumer Nodes

At the prompt:

consumer>

Enter consumer details in the format:

node priority demand

- **Lower priority number = higher importance**
- Enter **one consumer per line**
- Type done after entering all consumers

Example:

consumer> HOSP 1 8

consumer> RES 3 6

consumer> done

5. Enter Total Supply

When prompted:

Total supply value:

Enter the total available water supply.

Example:

Total supply value: 12

6. Program Execution

After completing the inputs, the program will automatically:

- Simulate **hourly snapshots**
 - Apply **sensor-based controls**
 - Predict consumer demands
 - Allocate water based on **priority-aware flow**
 - Display **graphs and outputs** from all connected modules
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Important Notes

- Always use **Source** as the source node name
 - All demand (consumer) nodes **must have a path to Sink**
 - If a path to Sink is missing, flow **cannot reach that node**
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